Evaluating Automatic CV Shortlisting Tool For Job Recruitment Based On Machine Learning Techniques

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***Abstract*—Recruitment and screening techniques have minimal limitations, which keeps candidates and HR from meeting expectations. Finding the finest candidates for a position by examining through hundreds of resumes requires a while and can introduce prejudice. These days, a lot of businesses use internet-based platforms to find new employees. These platforms, commonly referred to as "job portals," make the hiring process easier for both the recruiter and the candidate. The criteria can be established by HR based on factors like education, experience, and talents. This appears to have a significant impact on task reduction. However, there are still a large number of resumes that meet the criteria and must be manually reviewed. We propose in this paper, a tool for automatically shortlisting and ranking candidates based on their job profiles. By using cosine similarity to evaluate a resume against the job description, it streamlines the application process and makes it simple for the HR department to find the qualified applicant.**

***Keywords*—** CV shortlisting, resume, classification, machine learning, Cosine similarity

I. Introduction

In any market, talent acquisition is a difficult and time-consuming task for HR, which entails searching for the most relevant resumes against a job description. Finding a qualified candidate for a position among thousands of applicants is frequently a difficult task for a company [1]. Numerous individuals graduate from universities every six months. When these people apply for the jobs they want, hundreds of resumes are submitted for each open position. This causes a huge number of applications to be submitted, which presents a dilemma for the HR department. An effective online recruitment system can only help in this disastrous situation. Furthermore, HR personnel's lack of domain knowledge may make it difficult to select the best candidate for the job. Unsuitable or unfit candidates may not only have a negative impact on the company but may also result in the loss of a deserving candidate [2]. Many businesses are now using online recruitment systems to shortlist candidates based on criteria such as CGPA, degree, salary, experience, and so on. There is still a requirement for an automated system that can shortlist and rank candidates all at once. The amount of paperwork and human mistake would be greatly reduced with online resume classification. The classification and verification of resumes is the first step in the hiring process. Automating the first stage will greatly speed up the interview process for choosing candidates [2].

II. Literature Review

A. Problem Background

Finding and short-listing candidates for each post is typically a difficult and time-consuming process for HR professionals because it requires a lot of manual labour. Hundreds of resumes must be carefully sorted through and compared to the job specifications in order to narrow the field of candidates. Because resumes contain a lot of unstructured data, finding the material and correctly categorising it is the most difficult phase.

The format, size, and job scope of resumes vary greatly from one another, which makes the mismatch a very difficult issue. Both businesses and job seekers can speed up the hiring process with the aid of online recruiting tools [3]. Employers can screen individuals based on factors including talents, years of experience, education, and other factors. This might speed up the process, but there might still be a lot of resumes that fit the bill and need to be personally reviewed.

The research focused on resumes rank follows, almost the same approach. Data is collected from different websites and then pre-processed, transformed, and then compared with the keyword stored in the knowledge base to compute the similarity. Lexical analysis, syntactic analysis, semantic analysis morphological analysis, etc. are performed to achieve the results [10], [11].
Clustering-based methods are also used for resume classification purposes [12], [2]. Clustering is a process of creating groups based on similarities and differences. Each cluster is a collection of like-minded objects. Resume recommendations can also be taken as a classification problem. The model has been designed to categorize the resume as a web developer, accountant, HR, etc. [9]. Classification algorithms like KNN, Random Forest, Logistic Regression, and SVM can be used for classification [2].

B. Limitations of Existing Systems

For the purpose of separating data from resumes, many re-acquisition devices have been developed in recent years [13]. Although there are basic theories and preparation techniques for ex-

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TABLE I
ML APPROACHES USED IN LITERATURE

S.NO	*TITLE*	*Machine Learning Method*	*Category*
1	A ML approach for automation of resume Recommendation [4]	SWM, KNN	Recommendation
2	Automated CV classification using Clustering Technique [5]	K-Means, Clustering	Clustering
3	Resume information Extraction Framework [6]	DB scan	Classification, Clustering
4	Resume Parser with NLP [7]	Lexical analyzer	Lexical analyzer
5	Resume Ranking using NLP and ML [8]	Ranking Algorithm	Ranking
6	Intelligent Hiring with Resume Parser and Ranking [9]	Ranking Algorithm	Ranking
7	Online Resume Parser System Using Text Analysis [10]	Ranking Algorithm	Ranking
8	Resume Classification using various Machine Learning Algorithms [2]	SVM, Naive Bayes, Random Forest	Classification
9	Resume classification system using natural language processing and machine learning techniques [1]	SVM (Linear, SGD, SVC and NuSVC), Naive Bayes (Bernoulli, Multinomial and Gaussian), K-Nearest Neighbor (KNN), and Logistic Regression (LR)	Classification, Clustering

tracting information from the web, the majority of e-acquisition apparatuses still suffer from word handling and coordination, with reference to the requirements of the job. [14], [15]. A web-based tool is required to automatically shortlist and then rank the candidate based on their job profiles.

III. DESIGN AND METHODOLOGY

In this section, we have briefly described the process of gathering, pre-processing, and modeling the data for automated CV Shortlisting. We have also briefly explained the workflow and features of the HR portal.

A. Application Workflow and Features

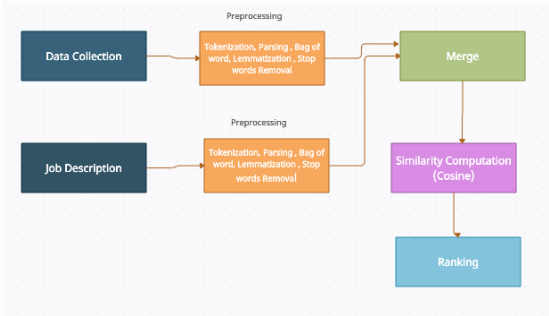


Fig. 1. Work Flow

B. Data Acquisition

We have collected data for LinkedIn using beautiful soap and selenium. From automatic login of LinkedIn we get username of different peoples, accessed their profiles and get text CV data using beautiful soap and save data in CSV file. A snippet of raw data is shown in Fig 2. We are accessing unlimited profiles in a single run of code. Raw data we are getting features are:

1) Name
2) Qualification
3) Institute Name
4) Previous/current Job post
5) Location
6) Title
7) Skills
8) Responsibilities (if any)

C. Data Processing

1) Bag of Words: To model text, using machine learning methods, the bag-of-words model is a way to represent text data. The raw text must be translated into well-defined fixed-length(vector) values before any algorithms can use it. For our dataset, Data processing is done by combining all content features in a single column to make a bag of words.

2) Lowercase Conversion: Even if terms like "SQA" and "sqa" have the same meaning when written in lower case, the vector space model represents them as two different words (resulting in more dimensions).

3) Stop Words Removal: One of the preprocessing techniques that is most frequently utilised across many NLP applications is stop word removal. The objective was to exclude words from our dataset that didn't contribute to the task's meaning.

4) Lemmatization: We applied the text normalisation technique known as lemmatization, which converts any sort of word to its base root mode. Lemmatization is responsible for combining several word inflections with the same meaning into their root forms. We utilise the nltk built-in library for lemmatization. Other than these steps we normalize the dataset by removing digits from all instances to make words clear.

D. Similarity Computation

We have job descriptions and candidate data. Fig 2 shows a sample job description taken from a website. Preprocessing is performed on both and then we get a set of keywords from the candidate profile and job description, respectively. These keywords are matched using cosine similarity metrics shown in 1.

$$\cos(\mathbf{x},\mathbf{y}) = \frac{\mathbf{x}\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{\sum_{i=1}^{n}\mathbf{x}_i\mathbf{y}_i}{\sqrt{\sum_{i=1}^{n}(\mathbf{x}_i)^2}\sqrt{\sum_{i=1}^{n}(\mathbf{y}_i)^2}} \quad (1)$$

Cosine similarity takes the point between two non-zero vectors and computes the cosine of that point, and this is known as a similarity. The score ranges from 0 to 1, 0 is the least similar, and 1 is the most similar.

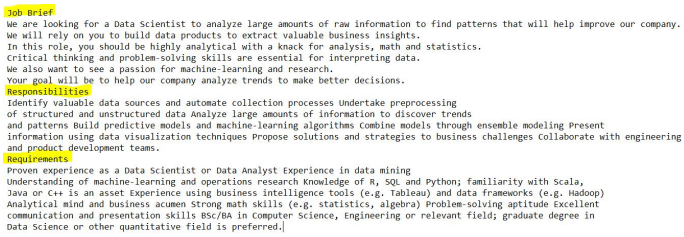


Fig. 2. Sample Job Description

E. Ranking Candidates

After getting the similarity of candidates against a job they are sorted in descending order of the computed cosine value. The highest cosine value means more similarity. We can pick top 'N' candidates from the list to shortlist for the job.

IV. MAJOR FEATURES OF THE APPLICATION

Application is divided into candidate and HR portals as shown in Fig 3, 4, 5 and 6. The candidate portal is used to create his/her profile and fill out the Personal Detail form, view vacancy Detail, and upload a CV in Fig 3 and 4. Candidates can also view the status of her application and their ranking among the candidates as well.

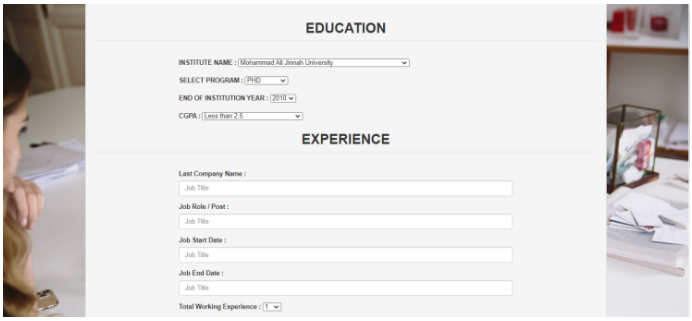


Fig. 3. Candidate Portal

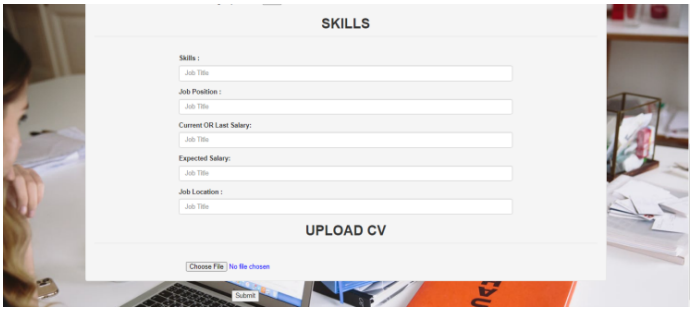


Fig. 4. Candidate Portal

HR portal is used to create a new job and set the selection criteria, download CV, filter candidates on criteria, and update requirements. Automatic shortlisting based on keywords matches with the given job description. This system can Rank the Top 'N' numbers of CVs in Fig 5 and 6.

V. EXPERIMENTAL SETUP AND RESULTS

We have taken 3 sample job descriptions for a Data scientist, SQA engineer, and Web developer. Each job description goes through the

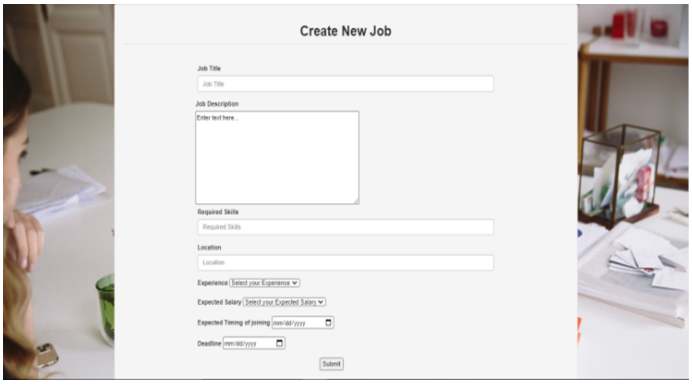


Fig. 5. HR Portal - Job Creation



Fig. 6. HR Portal - Short listing on criteria

data processing steps to get the keywords that can be matched with the candidate profile. Once preprocessing is done, we check all three jobs against the collected data from LinkedIn people's profiles and computed the similarity. Across all 222 Resumes, average of results ranges from 40.54% to 89.40% and count of Job description ranged from 1 to 1 the results average is shown in Fig 7 and Fig 8. We set the criteria to match if the similarity is greater than 70 percent We get 99 percent accuracy using cosine similarity against each job. Or in other words, all job descriptions matched against the correct job title. Table II shows the results.

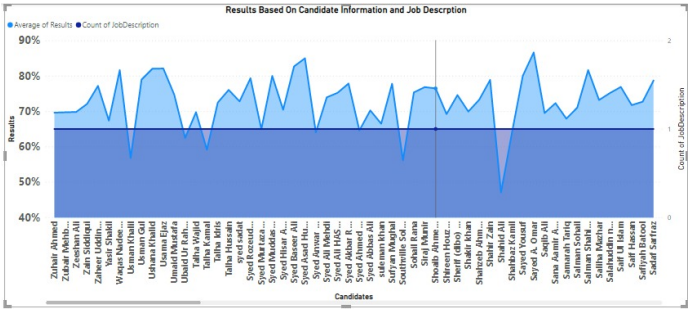


Fig. 7. Average Results based on Candidate information and Job Description

TABLE II
RESULT

S.No	*TITLE*	*Similarity*	*Actual*	*Predicted*
1	Data Scientist/Data Analyst	$\geq 70\%$	14	14
2	SQA Engineer	$\geq 70\%$	13	13
3	Web Development	$\geq 70\%$	49	49

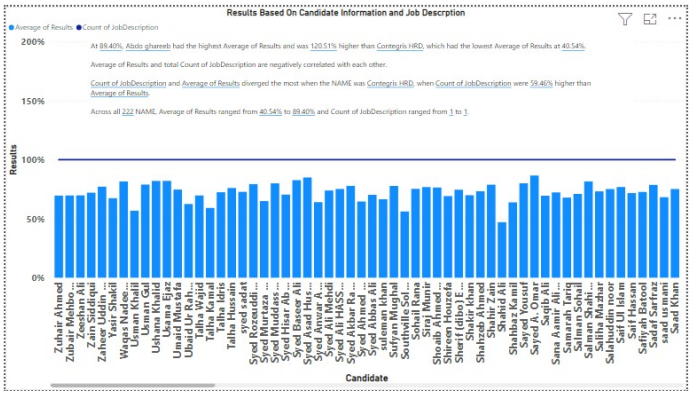


Fig. 8. Average Results based on Candidate information and Job Description

VI. CONCLUSION AND FUTURE WORK

In this research project, we have created a tool to automatically shortlist the candidates based on the similarity of candidates' profiles against the HR job description. In the future, we have planned to collect more data and create a huge dataset. Then we will work on the classification problem i.e., the system will predict which candidate is most suitable for which Job. Using this feature candidates enter their profiles, and the system will recommend the best job titles according to their profiles.

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